

Heterogeneous Parallel Key-Insulated Multi-Receiver Signcryption Scheme for IoV

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Abstract—The rapid growth of electric vehicle and autonomous vehicle populations has led to explosive expansion of IoV data being transmitted in the wireless communication infrastructure. Advances in IoV technologies also resulted in more complex and dynamic communication protocols/patterns, which are hard for the underlying wireless network to satisfy. Besides, security considerations of IoV communications require that key management must be stringently prohibit global failure mode of key management, meaning that, if a single IoV node compromises its private key, it will not lead to total security failure of the entire IoV network. To address these issues, in this paper, we propose a heterogeneous parallel key-insulated multi-receiver signcryption scheme for IoV (HPKI-MRSC). Firstly, the proposed scheme can realize one-to-many heterogeneous transmission, in which RSUs are deployed on certificateless cryptography (CLC) system, while vehicles are allocated in identity-based cryptography (IBC) system. In this manner, we observe that message transmission efficiency is improved greatly. Secondly, the parallel key-insulated mechanism can employ two helper keys to update private key periodically, and then solve key disclosure problem. Finally, when the number of receiver n is greater than or equal to 3, the proposed scheme has a lower signcryption overhead than other comparative schemes, and thus it is more suitable for IoV.

Index Terms—Heterogeneous, Parallel Key-Insulated, Signcryption, Internet of Vehicle (IoV).

I. INTRODUCTION

AS a superset of Vehicular Ad-hoc Networks (VANETs), Internet of Vehicle (IoV) pay more attention to information interaction among vehicles, organizations and Road Side Units (RSUs) [1]. With rapid development of IoV, users can obtain real-time road traffic information ubiquitously and improve travel comfort [2]. There exist various researches that focus on device communication allocated in a single cryptosystem [3]–[5]. Inevitably, in actual requirement, the devices of interaction sides are usually allocated in different cryptosystems. For example, due to different security requirements, when RSU and vehicles located in different cryptographic systems need to communicate with each other, it is an urgent research direction to construct a reasonable heterogeneous solution and satisfy different security requirements of interaction devices [6]–[8].

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Although IoV bring great advantages for individuals and organizations, it should be noted that the transmission of information often occurs in an untrusted channel during wireless networks, so it is important to ensure integrity, confidentiality, and unforgeability of information. Based on real-time requirements and security indicators, the primitive of signcryption is proposed in scheme [9]. This protocol can execute signature and encryption in a single step, which can greatly improve transmission efficiency [10]–[12]. Following this, Malone-Lee introduced an identity-based signcryption scheme (IBSC) [13], where the identity is considered as public key, and thus, the certificate management problem inherent in public key infrastructure (PKI) is solved. Then, more details based on IBSC are formulated such as [14]–[16]. Nonetheless, the private key of IBSC is generated by KGC, this however results in key escrow problem. Therefore, the construction of certificateless signcryption (CLSC) is proposed by Barbosa and Farshim [17]. In this manner, the certificate management problem and key escrow problem are both addressed.

However, the above schemes are oriented to a single cryptographic mechanism, thus it is not suitable for a heterogeneous environment where different transmission devices are govern under different cryptographic systems. Huang *et al.* [18] proposed a heterogeneous signcryption with key privacy function. In their scheme, message is delivered from identity-based cryptography (IBC) system to PKI system. Inspired by this mechanism, Li *et al.* [19] proposed two efficient signcryption schemes for heterogeneous systems. One scheme can transmit messages from PKI to IBC, and the other can transfer messages from IBC to PKI. Following this, a solution for messages delivered from certificateless cryptography (CLC) environment to PKI environment is constructed [20]. Recent years, more extensive research on heterogeneous signcryption has been provided specifically for actual IoV [21]–[23].

With the number of devices growing dramatically, if a message is delivered to n recipients by aforesaid schemes, it will perform n times one-to-one transmissions, which will bring a significant increase in signcryption computation and ciphertext communication overhead. In order to avoid this problem, a multi-receiver signcryption scheme that can realize one-to-many signcryption is presented. Sun and Li [24] discussed this problem, they introduced an efficient signcryption scheme between PKI and IBC with multi-receiver mechanism. Subsequently, Xie *et al.* [25] designed a heterogeneous multi-message multi-receiver scheme between IBC and CLC, to improve efficiency of sender during message-intensive system. However, once the signcryption private key is leaked, all data

protection mechanisms will lose the security [26].

In order to address the key exposure problem, Dodis *et al.* [27] first proposed a key-insulated public-key encryption scheme (KIPE). In this scheme, the public key of user remains unchanged in different time periods. Besides, this scheme introduces helper key and stores it in auxiliary device, which remains isolated from the network except for the time employed to update the private key. The finally generated temporary private key is utilized for signcryption operation. Nonetheless, the scheme in [27] only includes one helper key that needs to perform update operations continuously in a short period, this inevitably increases risk of private key contact with outside world. To solve this defect, a parallel key-insulated scheme is proposed in [26], where it allows two helper keys to update private key in sequence to reduce the exposure time of a single helper key. Therefore, this scheme can not only update sender's private key with period, but also can achieve a higher level of security. However, so far, there is no such parallel key-insulated scheme to avoid key exposure, and both realize one-to-many heterogeneous communication between vehicles simultaneously.

A. Our Motivation and Contributions

An imperative motivation encourages us to propose a secure scheme to address above concerns, such as key disclosure issue, one-to-many transmission problem and heterogeneous device security. In this paper, we propose a heterogeneous parallel key-insulated multi-receiver signcryption scheme for IoV (HPKI-MRSC). In this scheme, RSU focuses on the security of transmission, the vehicle focuses on real-time of transmission. The CLC cryptographic system is more secure than other cryptographic systems (such as PKI), and IBC system is more real-time than other cryptographic systems. Therefore, RSUs are more suitable for deployment on CLC system, while vehicles are more suitable for deployment on the IBC system. In this manner, the management, generation and storage of private keys are insulated from each other, to ensure that unauthorized access cannot occur. In addition, the parallel processing ability helps improve efficiency of this system. Even if one private key is compromised or cracked, the security of other private keys will not be affected. Moreover, the dynamic IoV environment requires low communication latency. The one-to-many data transmission mode in the proposed scheme makes information exchange more efficient, it takes less time for signcryption and is suitable for lightweight dynamic IoV devices. The concrete contributions are elaborated as below:

- *Heterogeneous-Communication*: Compared to other solutions simply considering single cryptosystem communication, our scheme takes into account the situation in which transmission devices are located in different cryptosystems. For instance, in case that the sender is allocated in a more secure CLC system, while multiple receivers are allocated in an IBC system, noticeably the proposed scheme provides a secure message transmission solution for the interaction of heterogeneous devices.
- *Efficiency-One-to-Many Mode*: Compared to previous one-to-one secure interactive modes, our HPKI-MRSC

scheme merges multi-receiver mechanism into the proposed construction, it allows RSU to signcrypt message and deliver ciphertext to multiple vehicles simultaneously. In this way, the efficiency of transmission is improved greatly with message-intensive environment.

- *Strong-Key-Insulated-Security*: Compared to traditional one-to-many secure communication modes, our HPKI-MRSC scheme combines parallel key-insulated method, it employs two helper keys to update the temporary private key during time period. Therefore, even if private key is leaked at a certain time, the temporary private key updated immediately at the next moment can prevent the attack caused by disclosure.

The related work is shown in section II. The relevant basic preliminaries are given in section III. In section IV, we elaborate the specific HPKI-MRSC construction. In section V, we prove the security of HPKI-MRSC via five theorems. In section VI, we analyze the performance comparison of schemes. In section VII, we summarize the conclusion of this paper.

II. RELATED WORKS

A. Parallel Key-Insulated Mechanism

Dodis *et al.* [27] first introduced key-insulated encryption scheme. Later, they proposed an extended key-insulated signature based on encryption [28]. Following this, the primitive of identity-based key-insulated signature (IBKIS) is constructed in [29], in this scheme, the identity is considered as the public key, and thus the certificate management problem is avoided. According to the observation, above schemes only include one helper key, when time period changes frequently, the temporary private key needs to be updated with the helper key in a short time simultaneously, and thus increases risk of key disclosure. To address this problem, Hanaoka *et al.* [26] proposed a parallel key-insulated scheme, in which two independent helper keys are utilized to generate the temporary private key, in this manner, it maintains time period update and ensures the security of private key concurrently. Based on this, a certificate-based parallel key-insulated aggregate signature is proposed [30], in this protocol, there exist four attacks from different adversaries that can be resisted. Yang *et al.* [31] constructed a pairing-free strong key-insulated certificateless signature, which generates or verifies authentication messages in a low-overhead way with security of private key. Then, a multiuser searchable encryption scheme with parallel key insulated function is introduced [32].

Compared the single cryptosystem reference [32], the first challenge of proposed scheme is to construct a multi-receiver signcryption scheme based on heterogeneous cryptosystems, the sender is located in CLC system with a higher level of security, and the receiver is located in IBC system with a higher efficiency requirement. Different cryptographic systems generate public/private key pairs in different ways, so it is difficult to design a reasonable heterogeneous multi-receiver signcryption scheme. The second challenge is to prove the security of proposed scheme. There exist three adversaries in the heterogeneous signcryption scheme from CLC to IBC.

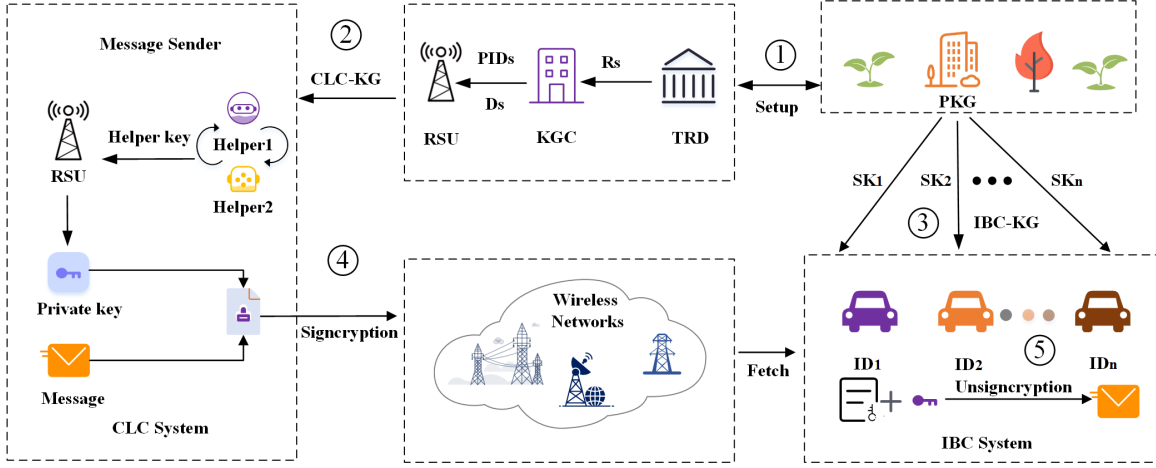


Fig. 1: The system model of HPKI-MRSC

Considering the parallel key insulated mechanism embedded in the CLC cryptosystem, we require to add two enhanced adversaries. Thus, the proposed scheme is attacked by employing five adversaries with different abilities. Then, the reasonable hard problems are embedded to prove the security of IND-CCA and EUF-CMA. At the same time, since the hard problems are unsolvable, it is reversely proved that the scheme cannot be breached accordingly. Therefore, the choice of adversaries, the choice of hard problem, the solution to address hard problem are both the challenge in this paper.

B. Multi-Receiver Signcryption

The concept of multi-receiver signcryption is proposed by Duan and Cao [33], which can signcrypt message and then broadcast it to all receivers. Some schemes based on multi-receiver mechanisms have been formulated [34]–[36]. However, the above schemes cannot guarantee anonymity, thus the receiver identity information can be easily obtained by attackers. Subsequently, in order to ensure anonymity of message, the protocol [37] proposes an anonymous multi-receiver encryption primitive, in this protocol, the identity information of all receivers is mixed into the Lagrange interpolation polynomial method to avoid the disclosure of identity. Later, several anonymous schemes based on this idea are gradually proposed [38], [39].

Certainly, the above schemes are just adapted to a single cryptosystem. Wang *et al.* [40] proposed a new efficient heterogeneous signcryption scheme that shifts between PKI and IBC. The proposed protocol is suitable for broadcast communication conditions in which senders can signcrypt multiple messages for multiple receivers simultaneously in the ad-hoc networks. Then, Qiu *et al.* [41] introduced a multi-receiver signcryption scheme for heterogeneous smart mobile IoT. Their scheme provides a communication way for heterogeneous devices from IBC to CLC. Besides, the bi-directional signcryption schemes for specific heterogeneous vehicle to infrastructure (V2I) communication are also proposed in [42], by allowing information broadcasting from PKI to multiple vehicles in IBC, or from IBC to multiple vehicles

in PKI. Nonetheless, the above literatures have not adequately achieved both the function of parallel key-insulated and multi-receiver signcryption.

TABLE I: Notions and symbols

Notations	Explanation
G_1, G_2	Two multiplication group
P	The generator of G_1
$H_i (0 \leq i \leq 6)$	Seven hash functions
λ	The security parameter
s_1, s_2	The system secret key
par	The system parameter
P_{pub}	The system public key
PK_s	The public key of sender
D_s	The partial private key of sender
$SK_{s,0}$	The initial full private key
$SK_{s,t}$	The temporary private key
sk_i	The private key of receiver
$ID_i, i = \{1, 2, 3, \dots, n\}$	The identity of receiver
C	The signcryption ciphertext
T_p	The bilinear pairing operation
E_1	The scalar multiplication in G_1
E_2	The exponentiation operation in G_2
T_G	The hash to G_1 operation

III. PRELIMINARY

In this section, the polynomial interpolation method, the bilinear groups, the protocol definition, the system and security model are demonstrated. The symbols that appear in this paper and their definitions are shown in Table I.

A. Lagrange Interpolating Polynomial Theorem

We define the following polynomial degree $n-1 \geq 0$, which

$$\sum_{i=1}^n F_i(x) = \sum_{i=1}^{n-1} a_i x^i$$

passes through n points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. For each i , we have:

$$F_i(x) = y_i \prod_{1 \leq j \neq i \leq n} \frac{x - x_j}{x_i - x_j}$$

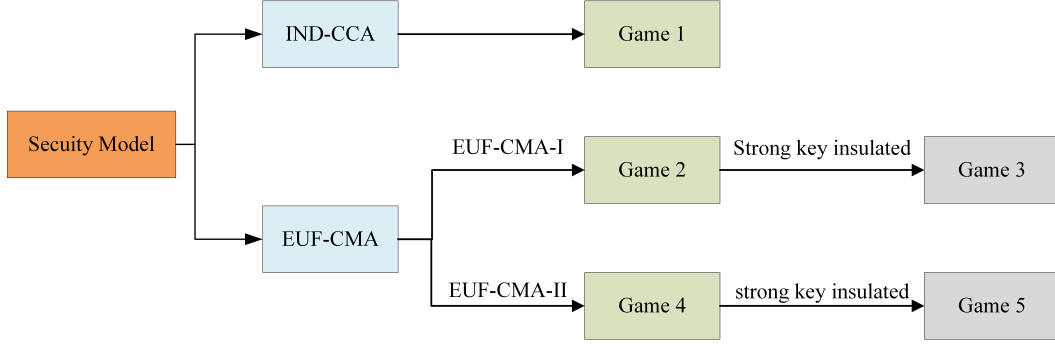


Fig. 2: The relationship among the different games

$$= \begin{cases} y_i, & x = x_i \\ 0, & x \in \{x_1, \dots, x_n\} - \{x_i\} \end{cases}$$

B. Bilinear Groups

We define two cyclic groups G_1 and G_2 with prime order q . Besides, the generator of G_1 is P . The bilinear map $e : G_1 \times G_1 \rightarrow G_2$ owns the properties as below:

- **Bilinearity:** $\forall a, d \in Z_p^*, \forall P, Q \in G_1, e(aP, dQ) = e(P, Q)^{ad}$.
- **Nondegeneracy:** $\exists P, Q \in G_1$ such that $e(P, Q) \neq 1$.
- **Computability:** $\forall P, Q \in G_1$, we can compute $e(P, Q)$.

C. Mathematical Problem

Bilinear Diffie-Hellman Problem (BDHP): Inputting an item (g, g^a, g^b, g^c) , the target of BDHP is to determine the equation $V \stackrel{?}{=} e(g, g)^{abc}$ holds or not, in which $a, b, c \in Z_p^*, V \in G_2$.

Bilinear Diffie-Hellman Assumption (BDHA): Suppose this assumption establishes, it refers that there is no probability polynomial time (PPT) adversary $\mathcal{A}$ can break the BDHP with a non-negligible advantage.

Computational Diffie-Hellman Problem (CDHP): Inputting an item $(g, g^a, g^b) \in G_1$, the target of CDHP is to compute the value of g^{ab} , where $a, b \in Z_p^*$.

Computational Diffie-Hellman Assumption (CDHA): Suppose this assumption establishes, there is no PPT adversary $\mathcal{A}$ can break the CDHP with a non-negligible advantage.

D. System Model

The system model of the proposed scheme is demonstrated in Fig. 1. There exist six entities to construct this model.

- **KGC:** It is in charge of generating the partial private key for RSU.
- **TRD:** The tamper resistance device is employed for generating the pseudonym for RSU.
- **Helper:** It mainly consists of two helpers, to interactively participate in updating the temporary private key of RSU.
- **RSU:** This entity is equipped on the CLC system, it is responsible for collecting information and transmitting it to multiple receiving vehicles in ciphertext.

- **PKG:** This entity mainly generates the private key of vehicles.
- **Vehicle:** This entity is allocated in the IBC system, where the authorized vehicle can receive ciphertext from RSU and recover message.

The trust relationship between PKG and KGC: In IBE cryptographic system, the public keys of user are usually derived from their identity information, PKG can generate the private key associated with identity, so PKG is a trusted third party. In CLC cryptographic system, KGC is also usually a trusted center that stores and manages the private key in the system. Besides, it also can distribute key generation, where KGC is responsible for generating partial private key. In this paper, PKG and KGC are considered trusted third parties and ensure that the generated keys are not compromised or abused. Other participants often rely on their security to ensure the overall security of the system.

Potential single point of failure: With the premise of PKG and KGC are trusted parties, in this paper, when data is transmitted from RSU located in the CLC system to vehicle located in the IBC system, RSU may encounter the single point of failure. Under the destruction of the attacker, the private key generated by RSU might be leaked and the ciphertext verification will fail. The key insulated mechanism in this scheme allows the helpers to update the private key at different time periods, and the current disclosure will not affect the private key in the next stage, which reasonably solves the problems of key leakage and transmission failure.

Overall flow: (1) KGC executes the Setup algorithm to generate the system parameter. (2) For RSU allocated in CLC system, when receiving the real identity of RSU, TRD generates the pseudonym for RSU. (3) When receiving the pseudonym of RSU, KGC generates the partial private key for RSU. (4) After receiving the partial private key, RSU employs two helpers to interactively generate and update the temporary private key of RSU. (5) When receiving the identities of vehicles allocated in IBC system, PKG generates the private keys of vehicles and delivers them to vehicles. (6) When intends to transmit a message M from RSU to vehicle, RSU executes the Signcryption operation to generate the ciphertext to multi-vehicle. (7) The receiver vehicle executes the Unsigncryption to recover the plaintext M .

E. Definition

- **Setup**(λ) $\rightarrow$ par : Taking security parameter λ as input, KGC generates the system parameter par .
- **CLC-Kengen**(par, ID_s) $\rightarrow$ ($SK_{s,t}, PK_s$): Taking par and the identity of sender ID_s as input, the following steps are conducted:
 - Pseudonym-Generate: The tamper resistance device outputs pseudonym of sender PID_s .
 - Public-Key-Generate: The sender outputs public key PK_s .
 - Partial-Private-Key-Generate: The KGC outputs partial private key D_s .
 - Initial-Private-Key-Generate: The sender outputs its initial private key $SK_{s,0}$.
 - Update*: The helper outputs update key $UK_{s,t}$ in period t .
 - Update: The sender outputs temporary private key $SK_{s,t}$ in period t .
- **IBC-Kengen**(par, ID_i) $\rightarrow$ SK_i : Taking par and the identity of receiver ID_i as input, PKG outputs private key of receiver SK_i .
- **Signcryption**($par, t, TK_t, M, SK_{s,t}, w$) $\rightarrow$ C : Given the input of M , the time period t , the private key of sender $SK_{s,t}$, the identity list of receiver $w = (ID_1, ID_2, ID_3, \dots, ID_n)$, where $ID_i \in w$, the sender executes this operation and outputs ciphertext C .
- **Unsigncryption**(par, C, SK_i, PK_s) $\rightarrow$ M or $\perp$: Given the input of ciphertext C , the private key of receiver SK_i , the public key of sender PK_s , the receiver $ID_i \in w$, $i = \{1, 2, 3, \dots, n\}$ outputs message M or $\perp$.

F. Security Model

Here, we set the semantic security of HPKI-MRSC via five games. The relationship among the different games is elaborated in Fig. 2.

Game 1: $\mathcal{A}_1$ and a challenger $\mathcal{C}$ conduct this game.

Setup: $\mathcal{C}$ conducts the **Setup** algorithm and generates the system parameter par to $\mathcal{A}_1$. Besides, $\mathcal{C}$ performs **CLC-Kengen** to output ($SK_{s,t}, PK_s$) to $\mathcal{A}_1$.

Phase 1: $\mathcal{A}_1$ conducts a series of queries as below.

- **Extract-Private-key-query:** Once receiving this query on ID_i , $\mathcal{C}$ returns SK_i to $\mathcal{A}_1$.
- **Unsigncryption-query:** Once receiving this query, the result of **Unsigncryption** (C, SK_i, PK_s) is delivered to $\mathcal{A}_1$.

Challenge: $\mathcal{A}_1$ generates two equal length plaintexts M_0 and M_1 , $\mathcal{C}$ picks a random bit $\beta \in \{0, 1\}$ and delivers $C^* = \text{Signcryption}(M_\beta, SK_{s,t}, w)$ to $\mathcal{A}_1$.

Phase 2: $\mathcal{A}_1$ performs the same operations as *Phase 1* with the following restrictions:

- $\mathcal{A}_1$ cannot execute the **Extract-Private-key-query** on $w^* = (ID_1^*, ID_2^*, ID_3^*, \dots, ID_n^*)$.
- $\mathcal{A}_1$ cannot execute the **Unsigncryption-query** on C^* .

Guess: $\mathcal{A}_1$ generates $\beta' \in \{0, 1\}$. If $\beta' = \beta$, $\mathcal{A}_1$ wins this game.

Definition 1: Suppose there is no adversary $\mathcal{A}_1$ with a non-negligible advantage wins this game, we say that the HPKI-MRSC can reach the IND-CCA security.

Game 2: $\mathcal{A}_2$ and a challenger $\mathcal{C}$ conduct this game. $\mathcal{A}_2$ cannot access the master secret key (s_1, s_2). However, it can execute the replace public key attack.

Setup: The **Setup** operation is conducted by $\mathcal{C}$ to output par . Moreover, the **IBC-Kengen** operation is also performed to output (SK_i, w) to $\mathcal{A}_2$.

Queries: $\mathcal{A}_2$ conducts a series of queries as below.

- **Extract-Partial-Private-key-query:** Once receiving this query on PID_s , $\mathcal{C}$ delivers D_s to $\mathcal{A}_2$.
- **Extract-Public-key-query:** Once receiving this query on PID_s , $\mathcal{C}$ delivers PK_s to $\mathcal{A}_2$.
- **Extract-Initial-Private-key-query:** Once receiving this query on PID_s , $\mathcal{C}$ delivers $SK_{s,0}$ to $\mathcal{A}_2$.
- **Extract-Temporary-Private-key-query:** Once receiving this query on (t, PID_s) , $\mathcal{C}$ delivers $SK_{s,t}$ to $\mathcal{A}_2$.
- **Replace-Public-key-query:** Once receiving this query, $\mathcal{C}$ sets $PK_s = PK'_s$ and delivers PK'_s to $\mathcal{A}_2$.
- **Signcryption-query:** Once receiving this query, $\mathcal{C}$ returns the $C^* = \text{Signcryption}(M, SK_{s,t}, w)$ to $\mathcal{A}_2$.

Forgery: After finishing the above queries, $\mathcal{A}_2$ generates two forged ciphertexts (t, C^*) and (t, C'^*) on PID_s^* . $\mathcal{A}_2$ wins this game if it satisfies the following requirements.

- $\mathcal{A}_2$ cannot access the **Extract-Partial-Private-key-query**, **Extract-Initial-Private-key-query** and **Extract-Temporary-Private-key-query** on PID_s .
- $\mathcal{A}_2$ cannot access the **Signcryption-query** on $(M, SK_{s,t}^*, w)$.
- The forged ciphertexts (t, C^*) and (t, C'^*) are valid.

Definition 2: Suppose there is no adversary $\mathcal{A}_2$ with a non-negligible advantage wins this game, we say that the HPKI-MRSC can reach the EUF-CMA-I security.

Game 3: $\mathcal{A}_3$ and a challenger $\mathcal{C}$ conduct this game. The ability of $\mathcal{A}_3$ is based on $\mathcal{A}_2$. This game is similar to Game 2, except that adversary $\mathcal{A}_3$ can perform the **Extract-Helper-Private-key-query** as follows:

- **Extract-Helper-Private-key-query:** Once receiving this query, $\mathcal{C}$ returns (hk_0, hk_1) to $\mathcal{A}_3$.

Definition 3: Suppose there is no adversary $\mathcal{A}_3$ with a non-negligible advantage wins this game, we say that the HPKI-MRSC can reach the strong key insulated EUF-CMA-I secure.

Game 4: $\mathcal{A}_4$ interacts with the challenger $\mathcal{C}$ to conduct this game. It can access the master secret key (s_1, s_2). However, it cannot execute the replace public key attack.

Setup: The **Setup** operation is conducted by $\mathcal{C}$ to output par . Moreover, the **IBC-Kengen** operation is also performed to output (SK_i, w) to $\mathcal{A}_4$.

Queries: $\mathcal{A}_4$ conducts a series of queries as below.

- The following queries are the same as Theorem 2, such as **Extract-Public-key-query**, **Extract-Initial-Private-key-query**, **Extract-Temporary-Private-key-query** and **Signcryption-query**.

Forgery: After finishing the above queries, $\mathcal{A}_4$ generates two forged ciphertexts (t, C^*) and (t, C'^*) on PID_s^* . If the

following requirements is satisfied, we say that $\mathcal{A}_4$ wins this game.

- $\mathcal{A}_4$ cannot access Extract-Initial-Private-key-query and Extract-Temporary-Private-key-query on PID_s .
- $\mathcal{A}_4$ cannot access the Signcryption-query on $(M, SK_{s,t}^*, w)$.
- The forged ciphertexts (t, C^*) and (t, C'^*) are valid.

Definition 4: Suppose there is no adversary $\mathcal{A}_4$ with a non-negligible advantage wins this game, we say that the HPKI-MRSC can reach the EUF-CMA-II security.

Game 5: $\mathcal{A}_5$ and a challenger $\mathcal{C}$ conduct this game. The ability of $\mathcal{A}_5$ is based on $\mathcal{A}_4$. This game is similar to Game 4, except that adversary $\mathcal{A}_5$ can perform the Extract-Helper-Private-key-query as follows:

- Extract-Helper-Private-key-query: Once receiving this query, $\mathcal{C}$ returns (hk_0, hk_1) to $\mathcal{A}_5$.

Definition 5: Suppose there is no adversary $\mathcal{A}_5$ with a non-negligible advantage wins this game, we say that the HPKI-MRSC can reach the strong key insulated EUF-CMA-II secure.

IV. CONSTRUCTION

In this section, we incorporate the parallel key insulated mechanism into the idea of heterogeneous multi-receiver signcryption. Firstly, the heterogeneous signcryption mechanism is suitable for the data transmission between two devices located in CLC and IBC. Then we employ the private key of the sender in CLC and the public key of the IBC receiver to signcrypt the data. Secondly, in order to reduce transmission redundancy, the multi-receiver concept means that a message is delivered from one sender to multiple recipients, to further reduce the transmission overhead. Finally, the parallel key insulated mechanism employs two helper keys to generate the final private key for signcryption in different time periods, this operation can avoid the potential key disclosure threat. The concrete constructed is demonstrated as follows.

- 1) **Setup(λ):** Given the security parameter λ , KGC picks the addition group G_1 and the multiplication group G_2 , the prime order of G_1 is q , the generator of G_1 is P , then it defines the bilinear pair $e : G_1 \times G_1 \rightarrow G_2$. Besides, it chooses six hash functions: $H_0 : G_1 \times G_1 \rightarrow \{0, 1\}^*$, $H_1 : \{0, 1\}^* \rightarrow G_1$, $H_2 : \{0, 1\}^* \times G_1 \times Z_p^* \rightarrow G_1$, $H_3 : \{0, 1\}^* \rightarrow G_1$, $H_4 : G_2 \times \{0, 1\}^* \times \{0, 1\}^* \times G_1 \rightarrow Z_p^*$, $H_5 : \{0, 1\}^n \times \{0, 1\}^* \times \{0, 1\}^* \times G_1 \times Z_p^n \rightarrow Z_p^*$, $H_6 : Z_p^* \times G_1^3 \rightarrow \{0, 1\}^n$. KGC selects $s_1 \in Z_p^*$, then computes $P_{pub} = s_1 P$. Besides, it selects two helper initial private keys $hk_0, hk_1 \in Z_p^*$, and calculates the helper initial public keys $TK_0 = hk_0 P$, $TK_1 = hk_1 P$. Meanwhile, PKG selects $s_2 \in Z_p^*$ and computes $T_{pub} = s_2 P$. The master secret key is (s_1, s_2) , the system public key is (P_{pub}, T_{pub}) . Finally, this algorithm publishes the system parameter $par = (G_1, G_2, q, P, e, P_{pub}, T_{pub}, TK_0, TK_1, H_0, H_1, H_2, H_3, H_4, H_5, H_6)$. The concrete system model is demonstrated in Fig. 3.

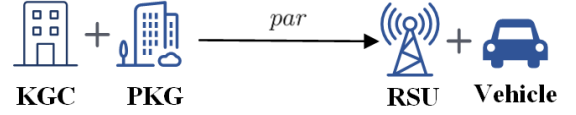


Fig. 3: The system model of Setup

- 2) **CLC-Kengen(par, ID_s):** The sender RSU is allocated in CLC system, this algorithm runs the following steps, the concrete system model is demonstrated in Fig. 4.
 - Pseudonym-Generate: After receiving the real identity ID_s , the tamper resistance device selects $r_s \in Z_p^*$, and computes $R_s = r_s \cdot P$, $PID_s = ID_s \oplus H_0(r_s \cdot P_{pub} || R_s)$. Then, it delivers R_s to the KGC via secure channel. Finally, it publishes the pseudonym PID_s .
 - Public-Key-Generate: The sender RSU picks $x_s \in Z_p^*$, then computes its public key $PK_s = x_s P$.
 - Partial-Private-Key-Generate: KGC first computes the partial private key $D_s = s_1 Q_s$, where $Q_s = H_1(PID_s)$. Then, it calculates two hash functions $f_{s,-1} = H_2(PID_s, TK_1, -1)$, $f_{s,0} = H_2(PID_s, TK_0, 0)$. Finally, KGC delivers $D_s, f_{s,-1}, f_{s,0}$ to RSU.
 - Initial-Private-Key-Generate: RSU computes its initial full private key $SK_{s,0} = D_s + (hk_1 + x_s) \cdot f_{s,-1} + hk_0 \cdot f_{s,0}$.
 - Update*: According to PID_s, par , the time period index t , and the i -th helper key hk_i , the i -th (where $i \equiv t \mod 2$) helper calculates $TK_i = hk_i \cdot P$ and generates update key of RSU: $UK_{s,t} = hk_i(f_{s,t} - f_{s,t-2})$.
 - Update: According to $UK_{s,t}$ and private key $SK_{s,t-1}$ for previous time period, RSU calculates temporary private key in time period t : $SK_{s,t} = SK_{s,t-1} + UK_{s,t} + x_s(f_{s,t-1} - f_{s,t-2})$. If $\varepsilon \equiv t \mod 2$, it computes $f_{s,t-1} = H_2(PID_s, TK_{\varepsilon'}, t-1)$. If $\varepsilon' \equiv (t-1) \mod 2$, it computes $f_{s,t} = H_2(PID_s, TK_{\varepsilon}, t)$. The temporary private key of RSU is $SK_{s,t} = D_s + (hk_{\varepsilon'} + x_s) \cdot f_{s,t-1} + hk_{\varepsilon} \cdot f_{s,t}$.

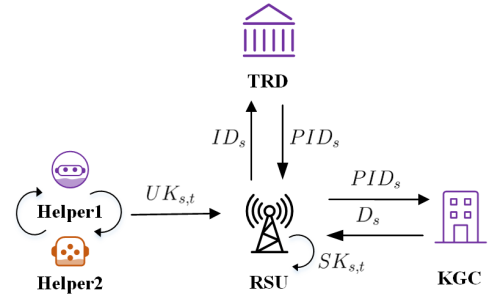


Fig. 4: The system model of CLC-Kengen

- 3) **IBC-Kengen(par, ID_i):** In this algorithm, the multiple vehicles are allocated in IBC system. PKG calculates the private key $SK_i = s_2 Q_i$ and delivers it to corresponding vehicle, where $Q_i = H_3(ID_i)$, $i = \{1, 2, 3, \dots, n\}$. The concrete system model is demonstrated in Fig. 5.

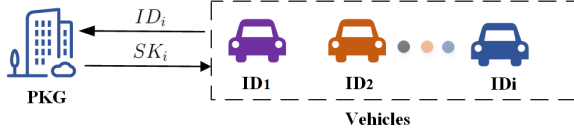


Fig. 5: The system model of IBC-Kengen

- 4) **Signcryption**($par, t, TK_t, M, SK_{s,t}, w$): Given the message M , the time period t , the secret key x_s , the private key of sender $SK_{s,t}$, the identity list of vehicle $w = (ID_1, ID_2, ID_3, \dots, ID_n)$ as input, where $ID_i \in w$, one RSU executes the following operations, the concrete system model is demonstrated in Fig. 6.

- Choose $k_1, k_2 \in Z_p^*$.
- Compute $C_1 = k_1 P$, $f_{s,t} = H_2(PID_s, TK_t, t)$, $\xi_i = e(T_{pub}, Q_i)^{x_s}$.
- Calculate $y_i = H_4(\xi_i, PID_s, ID_i, C_1)$ for each vehicle, where $i = \{1, 2, 3, \dots, n\}$.
- Choose $a_i \in Z_p^*$ and set $v = \{a_i | i = 0, \dots, n-1\}$. Then, calculate

$$f(x) = \prod_{i=1}^n (x - y_i) + k_2 \pmod{p}$$

$$= x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$$
- Compute $h = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$.
- Compute $C_2 = k_1 \cdot f_{s,t} + h \cdot SK_{s,t}$.
- Compute $C_3 = M \oplus H_6(k_2 || P_{pub} || C_1 || C_2)$.
- Generate the ciphertext $C = (C_1, C_2, C_3, v)$. Then it delivers C and w to the receiver.

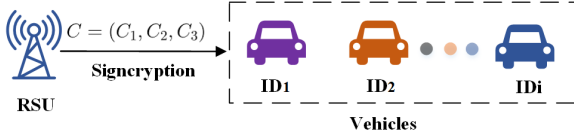


Fig. 6: The system model of Signcryption

- 5) **Unsigncryption**(par, C, SK_i, PK_s): Given the ciphertext C , the private key of vehicle SK_i , the public key of sender PK_s as input. The received vehicle $ID_i \in w$, $i = \{1, 2, 3, \dots, n\}$ executes the following operations, the concrete system model is demonstrated in Fig. 7.

- Calculate $\xi_i = e(PK_s, SK_i)$.
- Calculate $y_i = H_4(\xi_i, PID_s, ID_i, C_1)$.
- Calculate $k_2 = \sum_{i=0}^{n-1} a_i(y_i^i) + y_i^n \pmod{p}$.
- Calculate $M = C_3 \oplus H_6(k_2 || P_{pub} || C_1 || C_2)$.
- Calculate $h = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$.
- Check the following equation $e(C_2, P) \stackrel{?}{=} e(P_{pub}, Q_s)^h e(TK_{\epsilon'} + PK_s, f_{s,t-1})^h e(hTK_{\epsilon} + C_1, f_{s,t})$ holds or not, if holds, it returns message M' ; Else, it generates $\perp$.

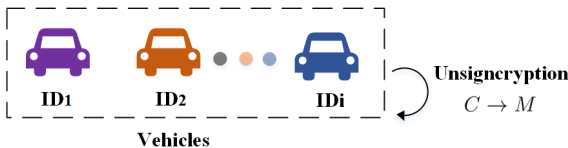


Fig. 7: The system model of Unsigncryption

Correctness: The involved equation is elaborated as below:

$$\begin{aligned} \xi_i &= e(PK_s, SK_i) \\ &= e(x_s P, s_2 Q_i) \\ &= e(s_2 P, Q_i)^{x_s} \\ &= e(T_{pub}, Q_i)^{x_s} \end{aligned}$$

$$\begin{aligned} e(C_2, P) &= e(k_1 \cdot f_{s,t} + h \cdot SK_{s,t}, P) \\ &= e(k_1 \cdot f_{s,t}, P) e(h \cdot SK_{s,t}, P) \\ &= e(C_1, f_{s,t}) e(SK_{s,t}, P)^h \\ &= e(C_1, f_{s,t}) e(D_s + (hk_{\epsilon'} + x_s) \cdot f_{s,t-1} + hk_{\epsilon} \cdot f_{s,t}, P)^h \\ &= e(C_1, f_{s,t}) e(P_{pub}, Q_s)^h e(TK_{\epsilon'} + PK_s, f_{s,t-1})^h \\ &\quad \cdot e(hTK_{\epsilon}, f_{s,t}) \\ &= e(P_{pub}, Q_s)^h e(TK_{\epsilon'} + PK_s, f_{s,t-1})^h \\ &\quad \cdot e(h \cdot TK_{\epsilon} + C_1, f_{s,t}) \end{aligned}$$

V. SECURITY ANALYSIS

In this section, we employ five theorems to prove the security of HPKI-MRSC.

Theorem 1: Assuming an adversary $\mathcal{A}_1$ can break the security of IND-CCA, that is to say, it has a challenger $\mathcal{C}$ that can address BDHP with a non-negligible advantage.

Proof: Given the input $(P, aP, bP, cP, X \in G_2)$, the objective of $\mathcal{C}$ is to determine whether $X = e(P, P)^{abc}$. The challenger $\mathcal{C}$ interacts with adversary $\mathcal{A}_1$ to execute the following steps.

Setup: $\mathcal{C}$ executes **Setup** algorithm and sets $T_{pub} = aP$, where $s_2 = a$, then it generates par to $\mathcal{A}_1$. Besides, $\mathcal{C}$ runs **CLC-Kengen** to deliver $(SK_{s,t}, PK_s)$ to $\mathcal{A}_1$, where $PK_s = bP$. Then, $\mathcal{A}_1$ outputs n target receivers $w^* = (ID_1^*, ID_2^*, ID_3^*, \dots, ID_n^*)$.

Phase 1: $\mathcal{A}_1$ adaptively conducts the following queries, $\mathcal{C}$ contains seven lists $\mathcal{L}_0, \mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6$ to response these queries.

- H_0 query: $\mathcal{C}$ sets the list $\mathcal{L}_0$ with the format of (r_s, P_{pub}, R_s, h_0) . When $\mathcal{A}_1$ queries $H_0(r_s \cdot P_{pub} || R_s)$, $\mathcal{C}$ randomly selects $h_0 \in \{0, 1\}^*$ and sets $h_0 = H_0(r_s \cdot P_{pub} || R_s)$, the item (r_s, P_{pub}, R_s, h_0) is inserted in $\mathcal{L}_0$. Then, $\mathcal{C}$ returns h_0 to $\mathcal{A}_1$.
- H_1 query: $\mathcal{C}$ sets the list $\mathcal{L}_1$ with the format of (PID_s, h_1) . When $\mathcal{A}_1$ queries $H_1(PID_s)$, $\mathcal{C}$ randomly selects $h_1 \in G_1$ and sets $h_1 = H_1(PID_s)$, the item (PID_s, h_1) is inserted in $\mathcal{L}_1$. Then, $\mathcal{C}$ returns h_1 to $\mathcal{A}_1$.
- H_2 query: $\mathcal{C}$ sets the list $\mathcal{L}_2$ with the format of $(PID_s, TK_{\epsilon}, TK_{\epsilon'}, \gamma_1, \gamma_2, U, U')$. $\mathcal{C}$ randomly chooses $\gamma_1, \gamma_2 \in Z_p^*$ and sets $U = f_{s,t} = H_2(PID_s, TK_{\epsilon}, t) = \gamma_1 P, U' = f_{s,t-1} = H_2(PID_s, TK_{\epsilon'}, t-1) = \gamma_2 P$. Then, the item $(PID_s, TK_{\epsilon}, TK_{\epsilon'}, \gamma_1, \gamma_2, U, U')$ is inserted in $\mathcal{L}_2$ and U, U' are returned to $\mathcal{A}_1$.
- H_3 query: $\mathcal{C}$ sets the list $\mathcal{L}_3$ with the format of (ID_i, h_3) . When $\mathcal{A}_1$ queries $H_3(ID_i)$, it is divided into two situations:

- If $ID_i \notin w^*$, $\mathcal{C}$ selects $\varphi \in Z_p^*$ and sets $h_3 = \varphi P$, the item (ID_i, h_3) is inserted in $\mathcal{L}_3$. Then, $\mathcal{C}$ returns h_3 to $\mathcal{A}_1$.
- If $ID_i \in w^*$, $\mathcal{C}$ sets $h_3 = cP$, the item (ID_i, h_3) is inserted in $\mathcal{L}_3$. Then, $\mathcal{C}$ returns h_3 to $\mathcal{A}_1$.
- H_4 query: $\mathcal{C}$ sets the list $\mathcal{L}_4$ with the format of $(\xi_i, PID_s, ID_i, C_1, h_4)$. When $\mathcal{A}_1$ queries $H_4(\xi_i, PID_s, ID_i, C_1)$, $\mathcal{C}$ randomly selects $h_4 \in Z_p^*$ and sets $h_4 = H_4(\xi_i, PID_s, ID_i, C_1)$, then the item $(\xi_i, PID_s, ID_i, C_1, h_4)$ is inserted in $\mathcal{L}_4$ and h_4 is returned to $\mathcal{A}_1$.
- H_5 query: $\mathcal{C}$ sets the list $\mathcal{L}_5$ with the format of $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$. When $\mathcal{A}_1$ makes this query, $\mathcal{C}$ randomly selects $h_5 \in Z_p^*$ and sets $h_5 = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$, then the item $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$ is inserted in $\mathcal{L}_5$ and h_5 is returned to $\mathcal{A}_1$.
- H_6 query: $\mathcal{C}$ sets the list $\mathcal{L}_6$ with the format of $(k_2, P_{pub}, C_1, C_2, h_6)$. When $\mathcal{A}_1$ queries $H_5(k_2 || P_{pub} || C_1 || C_2)$, $\mathcal{C}$ selects $h_6 \in \{0, 1\}^n$ and sets $h_6 = H_6(k_2 || P_{pub} || C_1 || C_2)$, then the item $(k_2, P_{pub}, C_1, C_2, h_6)$ is inserted in $\mathcal{L}_6$ and h_6 is returned to $\mathcal{A}_1$.
- Extract-Private-key-query: This reflects the case if receiving this query on ID_i , $\mathcal{C}$ conducts the following operations.
 - If $ID_i \notin w^*$, $\mathcal{C}$ returns $SK_i = \varphi aP$ to $\mathcal{A}_1$.
 - If $ID_i \in w^*$, $\mathcal{C}$ aborts this game.
- Unsigncryption-query: This reflects the case if receiving this query on ciphertext C , $\mathcal{C}$ firstly executes the Extract-Private-key-query on ID_i .
 - If $ID_i \notin w^*$, $\mathcal{C}$ returns the value of Unsigncryption (C, SK_i, PK_s) to $\mathcal{A}_1$.
 - If $ID_i \in w^*$, $\mathcal{C}$ retrieves $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$ and $(k_2, P_{pub}, C_1, C_2, h_6)$ from $\mathcal{L}_5$ and $\mathcal{L}_6$. Then, $\mathcal{C}$ does the following steps.
 - 1) Compute $C_3 \oplus h_6 = M'$.
 - 2) Compute $h = H_5(M', PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$.
 - 3) Check $e(C_2, P) \stackrel{?}{=} e(P_{pub}, Q_s)^h e(TK_{\varepsilon'} + PK_s, f_{s,t-1})^h e(h \cdot TK_{\varepsilon} + C_1, f_{s,t})$ and $h \stackrel{?}{=} h_5$ hold or not. Suppose these equation both hold, $\mathcal{C}$ will return M' to $\mathcal{A}_1$.

If the corresponding tuples are not found in the above lists, $\mathcal{C}$ will return $\perp$ to $\mathcal{A}_1$.

Challenge: $\mathcal{A}_1$ selects two messages M_0^*, M_1^* with equal length and a challenge receiver's identity ID_i to $\mathcal{C}$.

- If $ID_i \notin w^*$, $\mathcal{C}$ aborts.
- If $ID_i \in w^*$, $\mathcal{C}$ searches h_6^* from $\mathcal{L}_6$, then it chooses $\rho \in (0, 1)$ and $C_1^*, C_2^* \in G_1$.
 - Compute $h^* = H_5(M_\rho^*, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$.
 - Compute $C_3^* = M_\rho^* \oplus h_6^*$.
 - Finally, the ciphertext $C^* = (C_1^*, C_2^*, C_3^*, v, w)$ is returned to $\mathcal{A}_1$.

Phase 2: $\mathcal{A}_1$ executes the same queries as **Phase 1**. Nonetheless, $\mathcal{A}_1$ cannot execute the Extract-Private-key-query on $w^* = (ID_1^*, ID_2^*, ID_3^*, \dots, ID_n^*)$. Besides, it also cannot

execute the Unsigncryption-query on C^* .

Guess: $\mathcal{A}_1$ guesses the value of ρ' , if $\rho' = \rho$, $\mathcal{A}_1$ wins this game. Then, $\mathcal{C}$ searches the item $(\xi_i, PID_s, ID_i, C_1, h_4)$ from $\mathcal{L}_4$. Therefore, $\mathcal{C}$ can obtain $\xi_i = e(aP, cP)^b = e(P, P)^{abc}$.

Theorem 2: Suppose $\mathcal{A}_2$ can resist the EUF-CMA-I secure, there exists a challenger $\mathcal{C}$ that can address CDHP.

Proof: Given the input (P, aP, bP) , the objective of $\mathcal{C}$ is to compute abP . Then, $\mathcal{C}$ interacts with adversary $\mathcal{A}_2$ to conduct the following steps.

Setup: $\mathcal{C}$ performs **Setup** step, it sets $P_{pub} = aP$ and delivers par to $\mathcal{A}_2$, where $s_1 = a$. Besides, $\mathcal{C}$ runs IBC-Kengen to deliver (SK_i, w) to $\mathcal{A}_2$.

Queries: $\mathcal{A}_2$ adaptively conducts the following queries:

- H_0 query: $\mathcal{C}$ sets the list $\mathcal{L}_0$ with the format of (r_s, P_{pub}, R_s, h_0) . When $\mathcal{A}_2$ queries $H_0(r_s \cdot P_{pub} || R_s)$, $\mathcal{C}$ randomly selects $h_0 \in \{0, 1\}^*$ and sets $h_0 = H_0(r_s \cdot P_{pub} || R_s)$, the item (r_s, P_{pub}, R_s, h_0) is inserted in $\mathcal{L}_0$. Then, $\mathcal{C}$ returns h_0 to $\mathcal{A}_2$.
- H_1 query: When $\mathcal{A}_2$ queries $H_1(PID_s)$, it is divided into two situations:
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ selects $\gamma \in Z_p^*$ and sets $h_1 = \gamma P$, the item (ID_i, h_1) is inserted in $\mathcal{L}_1$. Then, $\mathcal{C}$ returns h_1 to $\mathcal{A}_2$.
 - If $PID_s = PID_s^*$, $\mathcal{C}$ sets $h_1 = \gamma bP$, the item (ID_i, h_1) is inserted in $\mathcal{L}_1$. Then, $\mathcal{C}$ returns h_1 to $\mathcal{A}_2$.
- H_2 query: $\mathcal{C}$ sets the list $\mathcal{L}_2$ with the format of $(PID_s, TK_{\varepsilon}, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$. $\mathcal{C}$ randomly chooses $\gamma_1, \gamma_2 \in Z_p^*$ and sets $U = f_{s,t} = H_2(PID_s, TK_{\varepsilon}, t) = \gamma_1 P, U' = f_{s,t-1} = H_2(PID_s, TK_{\varepsilon'}, t-1) = \gamma_2 P$. Then, the item $(PID_s, TK_{\varepsilon}, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$ is inserted in $\mathcal{L}_2$ and U, U' are returned to $\mathcal{A}_2$.
- H_3 query: $\mathcal{C}$ sets the list $\mathcal{L}_3$ with the format of (ID_i, h_3) . When $\mathcal{A}_2$ queries $H_3(ID_i)$, $\mathcal{C}$ randomly selects $h_3 \in G_1$ and sets $h_3 = H_1(ID_i)$, the item (ID_i, h_3) is inserted in $\mathcal{L}_3$. Then, $\mathcal{C}$ returns h_3 to $\mathcal{A}_2$.
- H_4 query: $\mathcal{C}$ sets the list $\mathcal{L}_4$ with the format of $(\xi_i, PID_s, ID_i, C_1, h_4)$. When $\mathcal{A}_2$ queries $H_4(\xi_i, PID_s, ID_i, C_1)$, $\mathcal{C}$ randomly selects $h_4 \in Z_p^*$ and sets $h_4 = H_4(\xi_i, PID_s, ID_i, C_1)$, then the item $(\xi_i, PID_s, ID_i, C_1, h_4)$ is inserted in $\mathcal{L}_4$ and h_4 is returned to $\mathcal{A}_2$.
- H_5 query: $\mathcal{C}$ sets the list $\mathcal{L}_5$ with the format of $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$. When $\mathcal{A}_2$ makes this query, $\mathcal{C}$ randomly selects $h_5 \in Z_p^*$ and sets $h_5 = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$, then the item $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$ is inserted in $\mathcal{L}_5$ and h_5 is returned to $\mathcal{A}_2$.
- H_6 query: $\mathcal{C}$ sets the list $\mathcal{L}_6$ with the format of $(k_2, P_{pub}, C_1, C_2, h_6)$. When $\mathcal{A}_2$ queries $H_5(k_2 || P_{pub} || C_1 || C_2)$, $\mathcal{C}$ selects $h_6 \in \{0, 1\}^n$ and sets $h_6 = H_6(k_2 || P_{pub} || C_1 || C_2)$, then the item $(k_2, P_{pub}, C_1, C_2, h_6)$ is inserted in $\mathcal{L}_6$ and h_6 is returned to $\mathcal{A}_2$.
- Extract-Partial-Private-key-query: This reflects the case if receiving this query, $\mathcal{C}$ sets a list $\mathcal{L}_{sk}$ and conducts the operation as below.
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ generates $D_s = \gamma aP = \gamma P_{pub}$

to $\mathcal{A}_2$. Finally, $\mathcal{C}$ inserts $(PID_s, D_s, x_s, PK_s, SK_{s,0})$ to $\mathcal{L}_{sk}$ and delivers D_s to $\mathcal{A}_2$.

- If $PID_s = PID_s^*$, $\mathcal{C}$ aborts this game. Finally, $\mathcal{C}$ inserts $(PID_s, \perp, x_s, PK_s, \perp)$ to $\mathcal{L}_{sk}$.

- Extract-Public-key-query: This reflects the case if receiving this query, $\mathcal{C}$ first searches whether the $\mathcal{L}_{sk}$ contains PID_s , if exists, $\mathcal{C}$ delivers PK_s to $\mathcal{A}_2$. If not exist, it selects $x_s \in Z_p^*$, computes $PK_s = x_s P$, and inserts $(PID_s, D_s, x_s, PK_s, SK_{s,0})$ to $\mathcal{L}_{sk}$. Finally, $\mathcal{C}$ delivers PK_s to $\mathcal{A}_2$.

- Extract-Initial-Private-key-query: This reflects the case if receiving this query on PID_s , $\mathcal{C}$ first searches whether the $\mathcal{L}_{sk}$ contains PID_s .

- If $PID_s \neq PID_s^*$, $\mathcal{C}$ picks $hk_0, hk_1 \in Z_p^*$, then it computes $SK_{s,0} = D_s + (hk_1 + x_s) \cdot U' + hk_0 \cdot U$ and delivers $SK_{s,0}$ to $\mathcal{A}_2$.
- If $PID_s = PID_s^*$, $\mathcal{C}$ aborts this game.

- Extract-Temporary-Private-key-query: This reflects the case if receiving this query, $\mathcal{C}$ first searches whether the $\mathcal{L}_{sk}$ contains PID_s .

- If $PID_s \neq PID_s^*$, $\mathcal{C}$ executes the following steps.

- 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $SK_{s,t} = D_s + (hk_0 + x_s) \cdot U' + hk_1 \cdot U$. Therefore, $SK_{s,t} = D_s + \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot TK_1$.
- 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $SK_{s,t} = D_s + (hk_1 + x_s) \cdot U' + hk_0 \cdot U$. Therefore, $SK_{s,t} = D_s + \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot TK_0$.

- If $PID_s = PID_s^*$, $\mathcal{C}$ aborts this game.

- Replace-Public-key-query: After receiving this query, $\mathcal{C}$ checks $(PID_s, D_s, x_s, PK_s, SK_{s,0})$ from $\mathcal{L}_{sk}$, then it sets $PK_s = PK'_s$, $x_s = \perp$ and inserts $(PID_s, D_s, \perp, PK'_s, SK_{s,0})$ to $\mathcal{L}_{sk}$.

- Signcryption-query: After receiving this query on (PID_s, ID_i, M, t) , $\mathcal{C}$ first executes the H_1 query and H_2 query. It searches $\mathcal{L}_2$, $\mathcal{L}_6$ and $\mathcal{L}_{sk}$ for $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$, $(k_2, P_{pub}, C_1, C_2, h_6)$ and $(PID_s, D_s, x_s, PK_s, SK_{s,0})$. Besides, It picks $k_1 \in Z_p^*$ and computes $C_1 = k_1 P$, $C_3 = M \oplus h_6$. Then, it sets $h = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$.

- 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot (h \cdot TK_1 + C_1)$.

- 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot (h \cdot TK_0 + C_1)$.

Finally, $\mathcal{C}$ generates the valid ciphertext $C = (C_1, C_2, C_3, v, w)$ to $\mathcal{A}_2$.

Forgery: When $\mathcal{A}_2$ executes all the queries, according to the forgery theorem, it will generate two valid ciphertexts on PID_s^* , such as $C^* = (C_1^*, C_2^*, C_3^*, v^*, w^*)$ and $C'^* = (C_1'^*, C_2'^*, C_3'^*, v^*, w^*)$, from the above queries, we can obtain $P_{pub} = aP$, $Q_s^* = h_1 = \gamma bP$, $f_{s,t} = \gamma_1 P$, $f_{s,t-1} = \gamma_2 P$, $C_1^* = k_1 P$.

- 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \cdot \gamma \cdot abP + h \cdot \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot h \cdot TK_1 + \gamma_1 \cdot C_1^*.$$

$$C_2'^* = h' \cdot \gamma \cdot abP + h' \cdot \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot h' \cdot TK_1 + \gamma_1 \cdot C_1'^*.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot abP + \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot TK_1)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma_2 \cdot (TK_0 + PK_s) - \gamma_1 \cdot TK_1] \gamma^{-1}$$

- 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \cdot \gamma \cdot abP + h \cdot \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot h \cdot TK_0 + \gamma_1 \cdot C_1^*.$$

$$C_2'^* = h' \cdot \gamma \cdot abP + h' \cdot \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot h' \cdot TK_0 + \gamma_1 \cdot C_1'^*.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot abP + \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot TK_0)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma_2 \cdot (TK_1 + PK_s) - \gamma_1 \cdot TK_0] \gamma^{-1}$$

Theorem 3: Suppose $\mathcal{A}_3$ can achieve the strong key insulated EUF-CMA-I secure based on $\mathcal{A}_2$, there exists a challenger $\mathcal{C}$ that can address CDHP.

Proof: Given the input (P, aP, bP) , the objective of $\mathcal{C}$ is to compute abP . Then, $\mathcal{C}$ interacts with adversary $\mathcal{A}_3$ to conduct the following steps.

Setup: $\mathcal{C}$ performs Setup step, it sets $P_{pub} = aP$ and delivers par to $\mathcal{A}_3$, where $s_1 = a$. Besides, $\mathcal{C}$ runs IBC-Kengen to deliver (SK_i, w) to $\mathcal{A}_3$.

Queries: $\mathcal{A}_3$ adaptively conducts the following queries:

- These queries are the same as **Theorem 2**, such as H_i query ($0 \leq i \leq 6$), Extract-Partial-Private-key-query, Extract-Public-key-query, Extract-Initial-Private-key-query, Extract-Temporary-Private-key-query, Replace-Public-key-query.

- Extract-Helper-Private-key-query: After receiving this query, $\mathcal{C}$ picks $d_0, d_1 \in Z_p^*$, calculates $hk_0 = d_0$, $hk_1 = d_1$, $TK_0 = d_0 P$, $TK_1 = d_1 P$. Then, $\mathcal{C}$ maintains the helper list $\mathcal{L}_{hp} = \{hk_0, hk_1, TK_0, TK_1\}$.

- Signcryption-query: After receiving this query on (PID_s, M, t) , $\mathcal{C}$ first executes the H_1 query and H_2 query. Then it searches $\mathcal{L}_2$, $\mathcal{L}_6$ and $\mathcal{L}_{sk}$ for $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$, $(k_2, P_{pub}, C_1, C_2, h_6)$ and $(PID_s, D_s, x_s, PK_s, SK_{s,0})$. It picks $k_1 \in Z_p^*$ and computes $C_1 = k_1 P$, $C_3 = M \oplus h_6$.

- 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot U' \cdot (hk_0 + x_s) + U \cdot (h \cdot hk_1 + k_1)$.

- 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot U' \cdot (hk_1 + x_s) + U \cdot (h \cdot hk_0 + k_1)$.

Finally, $\mathcal{C}$ generates the valid ciphertext $C = (C_1, C_2, C_3, v, w)$ to $\mathcal{A}_3$.

Forgery: When $\mathcal{A}_3$ executes all the queries, according to the forgery theorem, it will generate two valid ciphertexts on PID_s^* , such as $C^* = (C_1^*, C_2^*, C_3^*, v^*, w^*)$ and $C'^* = (C_1'^*, C_2'^*, C_3'^*, v^*, w^*)$, from the above queries, we can obtain $P_{pub} = aP$, $Q_s^* = h_1 = \gamma bP$, $C_1^* = k_1 P$, $U = f_{s,t} = \gamma_1 P$, $U' = f_{s,t-1} = \gamma_2 P$, $C_1^* = k_1 P$.

- 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \cdot \gamma \cdot abP + h \cdot U' \cdot (hk_0 + x_s) + U \cdot h \cdot hk_1 + U \cdot k_1.$$

$$C_2'^* = h' \cdot \gamma \cdot abP + h' \cdot U' \cdot (hk_0 + x_s) + U \cdot h' \cdot hk_1 + U \cdot k_1.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot abP + U' \cdot (hk_0 + x_s) + U \cdot hk_1)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - U' \cdot (hk_0 + x_s) - U \cdot hk_1] \gamma^{-1}$$

2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \cdot \gamma \cdot abP + h \cdot U' \cdot (hk_1 + x_s) + U \cdot h \cdot hk_0 + U \cdot k_1.$$

$$C_2'^* = h' \cdot \gamma \cdot abP + h' \cdot U' \cdot (hk_1 + x_s) + U \cdot h' \cdot hk_0 + U \cdot k_1.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot abP + U' \cdot (hk_1 + x_s) + U \cdot hk_0)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - U' \cdot (hk_1 + x_s) - U \cdot hk_0] \gamma^{-1}$$

Theorem 4: Suppose $\mathcal{A}_4$ can resist the EUF-CMA-II secure, there exists a challenger $\mathcal{C}$ that can address CDHP.

Proof: Given the input (P, aP, bP) , the objective of $\mathcal{C}$ is to compute abP . Then, $\mathcal{C}$ interacts with adversary $\mathcal{A}_4$ to conduct the following steps.

Setup: $\mathcal{C}$ performs Setup step, it sets $P_{pub} = s_1P$ and delivers par to $\mathcal{A}_4$. Besides, $\mathcal{C}$ runs IBC-Kengen to deliver (SK_i, w) to $\mathcal{A}_4$.

Queries: $\mathcal{A}_4$ adaptively conducts the following queries:

- H_0 query: $\mathcal{C}$ sets the list $\mathcal{L}_0$ with the format of (r_s, P_{pub}, R_s, h_0) . When $\mathcal{A}_4$ queries $H_0(r_s \cdot P_{pub} || R_s)$, $\mathcal{C}$ randomly selects $h_0 \in \{0, 1\}^*$ and sets $h_0 = H_0(r_s \cdot P_{pub} || R_s)$, the item (r_s, P_{pub}, R_s, h_0) is inserted in $\mathcal{L}_0$. Then, $\mathcal{C}$ returns h_0 to $\mathcal{A}_4$.
- H_1 query: When $\mathcal{A}_4$ queries $H_1(PID_s)$, $\mathcal{C}$ selects $\gamma \in Z_p^*$ and sets $h_1 = \gamma P$, the item (ID_i, h_1) is inserted in $\mathcal{L}_1$. Then, $\mathcal{C}$ returns h_1 to $\mathcal{A}_4$.
- H_2 query: $\mathcal{C}$ sets the list $\mathcal{L}_2$ with the format of $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$.
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ selects $\gamma_1, \gamma_2 \in Z_p^*$, $TK_\varepsilon, TK_{\varepsilon'} \in G_1$ and sets $U = f_{s,t} = H_2(PID_s, TK_\varepsilon, t) = \gamma_1 P$, $U' = f_{s,t-1} = H_2(PID_s, TK_{\varepsilon'}, t-1) = \gamma_2 P$, the item $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$ is inserted in $\mathcal{L}_2$. Then, $\mathcal{C}$ returns U, U' to $\mathcal{A}_4$.
 - If $PID_s = PID_s^*$, $\mathcal{C}$ selects $\gamma_1, \gamma_2, \lambda_0, \lambda_1 \in Z_p^*$ and sets $U = f_{s,t} = H_2(PID_s, TK_\varepsilon, t) = \gamma_1 P$, $U' = f_{s,t-1} = H_2(PID_s, TK_{\varepsilon'}, t-1) = \gamma_2 bP$, $TK_\varepsilon = \lambda_0 P$, $TK_{\varepsilon'} = \lambda_1 P$. The item $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$ is inserted in $\mathcal{L}_2$. Then, $\mathcal{C}$ returns U, U' to $\mathcal{A}_4$.
- H_3 query: $\mathcal{C}$ sets the list $\mathcal{L}_3$ with the format of (ID_i, h_3) . When $\mathcal{A}_4$ queries $H_3(ID_i)$, $\mathcal{C}$ randomly selects $h_3 \in G_1$ and sets $h_3 = H_1(ID_i)$, the item (ID_i, h_3) is inserted in $\mathcal{L}_3$. Then, $\mathcal{C}$ returns h_3 to $\mathcal{A}_4$.
- H_4 query: $\mathcal{C}$ sets the list $\mathcal{L}_4$ with the format of $(\xi_i, PID_s, ID_i, C_1, h_4)$. When $\mathcal{A}_4$ queries $H_4(\xi_i, PID_s, ID_i, C_1)$, $\mathcal{C}$ randomly selects $h_4 \in Z_p^*$ and sets $h_4 = H_4(\xi_i, PID_s, ID_i, C_1)$, then the item

$(\xi_i, PID_s, ID_i, C_1, h_4)$ is inserted in $\mathcal{L}_4$ and h_4 is returned to $\mathcal{A}_4$.

- H_5 query: $\mathcal{C}$ sets the list $\mathcal{L}_5$ with the format of $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$. When $\mathcal{A}_4$ makes this query, $\mathcal{C}$ randomly selects $h_5 \in Z_p^*$ and sets $h_5 = H_5(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1})$, then the item $(M, PID_s, ID_i, C_1, a_0, \dots, a_{n-1}, h_5)$ is inserted in $\mathcal{L}_5$ and h_5 is returned to $\mathcal{A}_4$.
- H_6 query: $\mathcal{C}$ sets the list $\mathcal{L}_6$ with the format of $(k_2, P_{pub}, C_1, C_2, h_6)$. When $\mathcal{A}_4$ queries $H_5(k_2 || P_{pub} || C_1 || C_2)$, $\mathcal{C}$ selects $h_6 \in \{0, 1\}^n$ and sets $h_6 = H_6(k_2 || P_{pub} || C_1 || C_2)$, then the item $(k_2, P_{pub}, C_1, C_2, h_6)$ is inserted in $\mathcal{L}_6$ and h_6 is returned to $\mathcal{A}_4$.
- Extract-Public-key-query: This reflects the case if receiving this query, $\mathcal{C}$ conducts the following operations:
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ it selects $x_s \in Z_p^*$ and computes $PK_s = x_s P$ to $\mathcal{A}_4$. Finally, $\mathcal{C}$ inserts $(PID_s, D_s, x_s, PK_s, SK_{s,0})$ to $\mathcal{L}_{sk}$ and delivers PK_s to $\mathcal{A}_4$.
 - If $PID_s = PID_s^*$, $\mathcal{C}$ sets $PK_s = aP$ to $\mathcal{A}_4$. Finally, $\mathcal{C}$ inserts $(PID_s, D_s, \perp, PK_s, \perp)$ to $\mathcal{L}_{sk}$.
- Extract-Initial-Private-key-query: This reflects the case if receiving this query, $\mathcal{C}$ first searches whether the $\mathcal{L}_{sk}$ contains PID_s .
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ picks $hk_0, hk_1, x_s \in Z_p^*$, then it computes $SK_{s,0} = D_s + (hk_1 + x_s) \cdot U' + hk_0 \cdot U$ and delivers $SK_{s,0}$ to $\mathcal{A}_4$.
 - If $PID_s = PID_s^*$, $\mathcal{C}$ aborts this game.
- Extract-Temporary-Private-key-query: This reflects the case if receiving this query, $\mathcal{C}$ first searches whether the $\mathcal{L}_{sk}$ contains PID_s .
 - If $PID_s \neq PID_s^*$, $\mathcal{C}$ picks $x_s \in Z_p^*$ and executes the following steps.
 - 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $SK_{s,t} = D_s + (hk_0 + x_s) \cdot U' + hk_1 \cdot U$. Therefore, $SK_{s,t} = D_s + \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot TK_1$.
 - 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $SK_{s,t} = D_s + (hk_1 + x_s) \cdot U' + hk_0 \cdot U$. Therefore, $SK_{s,t} = D_s + \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot TK_0$.
 - If $PID_s = PID_s^*$, $\mathcal{C}$ aborts this game.
- Signcryption-query: After receiving this query on (PID_s, ID_i, M, t) , $\mathcal{C}$ first executes the H_1 query and H_2 query. Then it searches $\mathcal{L}_2$, $\mathcal{L}_6$ and $\mathcal{L}_{sk}$ for $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$, $(k_2, P_{pub}, C_1, C_2, h_6)$ and $(PID_s, D_s, x_s, PK_s, SK_{s,0})$. It picks $k_1 \in Z_p^*$ and computes $C_1 = k_1 P$, $C_3 = M \oplus h_6$.
 - 1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (TK_0 + PK_s) + \gamma_1 \cdot (h \cdot TK_1 + C_1)$.
 - 2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (TK_1 + PK_s) + \gamma_1 \cdot (h \cdot TK_0 + C_1)$.
 Finally, $\mathcal{C}$ generates the valid ciphertext $C = (C_1, C_2, C_3, v, w)$ to $\mathcal{A}_4$.

Forgery: When $\mathcal{A}_4$ executes all the queries, according to the forgery theorem, it will generate two valid ciphertexts on PID_s^* , such as $C^* = (C_1^*, C_2^*, C_3^*, v^*, w^*)$ and $C'^* =$

$(C_1^*, C_2^*, C_3^*, v^*, w^*)$, from the above queries, we can obtain $PK_s = aP$, $Q_s^* = h_1 = \gamma P$, $f_{s,t} = \gamma_1 P$, $f_{s,t-1} = \gamma_2 bP$, $TK_0 = \lambda_0 P$, $TK_1 = \lambda_1 P$, $C_1^* = k_1 P$.

1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$\begin{aligned} C_2^* &= h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (\lambda_0 bP + abP) + \gamma_1 \cdot h \cdot TK_1 + \gamma_1 \cdot C_1^* \\ C_2'^* &= h' \cdot \gamma \cdot P_{pub} + h' \cdot \gamma_2 \cdot (\lambda_0 bP + abP) + \gamma_1 \cdot h' \cdot TK_1 + \gamma_1 \cdot C_1^* \end{aligned}$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot P_{pub} + \gamma_2 \cdot (\lambda_0 bP + abP) + \gamma_1 \cdot TK_1)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma P_{pub} - \gamma_2 \lambda_0 bP - \gamma_1 TK_1] \gamma_2^{-1} \gamma_2 bP. TK_0 = hk_0 P, TK_1 = hk_1 P, C_1^* = k_1 P.$$

2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$\begin{aligned} C_2^* &= h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (\lambda_1 bP + abP) + \gamma_1 \cdot h \cdot TK_0 + \gamma_1 \cdot C_1^* \\ C_2'^* &= h' \cdot \gamma \cdot P_{pub} + h' \cdot \gamma_2 \cdot (\lambda_1 bP + abP) + \gamma_1 \cdot h' \cdot TK_0 + \gamma_1 \cdot C_1^* \end{aligned}$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot P_{pub} + \gamma_2 \cdot (\lambda_1 bP + abP) + \gamma_1 \cdot TK_0)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma P_{pub} - \gamma_2 \lambda_1 bP - \gamma_1 TK_0] \gamma_2^{-1} \gamma_2 bP.$$

TABLE II: The execution time of cryptographic operations

Operation	T_p	E_1	E_2	T_G
1-Time(ms)	0.6244	1.1950	0.0961	2.2400
2-Time(ms)	0.6393	0.9670	0.0989	2.1000
3-Time(ms)	0.5623	1.2660	0.0679	2.1629

Theorem 5: Suppose $\mathcal{A}_5$ can resist the strong key insulated EUF-CMA-II secure based on $\mathcal{A}_4$, there exists a challenger $\mathcal{C}$ that can address CDHP.

Proof: Given the input (P, aP, bP) , the objective of $\mathcal{C}$ is to compute abP . Then, $\mathcal{C}$ interacts with adversary $\mathcal{A}_5$ to conduct the following steps.

Setup: $\mathcal{C}$ performs Setup step, it sets $P_{pub} = s_1 P$ and delivers *par* to $\mathcal{A}_5$. Besides, $\mathcal{C}$ runs IBC-Kengen to deliver (SK_i, w) to $\mathcal{A}_5$.

Queries: $\mathcal{A}_5$ adaptively conducts the following queries:

- These queries are the same as **Theorem 4**, such as H_i query ($0 \leq i \leq 6$), Extract-Public-key-query, Extract-Initial-Private-key-query, Extract-Temporary-Private-key-query.
- Extract-Helper-Private-key-query: After receiving this query, $\mathcal{C}$ picks $\lambda_0, \lambda_1 \in Z_p^*$, calculates $hk_0 = \lambda_0$, $hk_1 = \lambda_1$, $TK_0 = hk_0 P = \lambda_0 P$, $TK_1 = hk_1 P = \lambda_1 P$. Then, $\mathcal{C}$ maintains the helper list $\mathcal{L}_{hp} = \{hk_0, hk_1, TK_0, TK_1\}$.
- Signcryption-query: After receiving this query on (PID_s, M, t) , $\mathcal{C}$ first executes the H_1 query and H_2 query. Then it searches $\mathcal{L}_2$, $\mathcal{L}_6$, $\mathcal{L}_{hp}$, $\mathcal{L}_{sk}$ for $(PID_s, TK_\varepsilon, TK_{\varepsilon'}, \gamma_1, \gamma_2, U, U')$, $(k_2, P_{pub}, C_1, C_2, h_6)$, (hk_0, hk_1, TK_0, TK_1) and $(PID_s, D_s, x_s, PK_s, SK_{s,0})$. It picks $k_1 \in Z_p^*$ and computes $C_1 = k_1 P$, $C_3 = M \oplus h_6$.

1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (hk_0 \cdot P + PK_s) + U \cdot (h \cdot hk_1 + k_1)$, where $PK_s = aP$.

2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes $C_2 = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 \cdot (hk_1 \cdot P + PK_s) + U \cdot (h \cdot hk_0 + k_1)$.

Finally, $\mathcal{C}$ generates the valid ciphertext $C = (C_1, C_2, C_3, v, w)$ to $\mathcal{A}_5$.

Forgery: When $\mathcal{A}_5$ executes all the queries, according to the forgery theorem, it will generate two valid ciphertexts on PID_s^* , such as $C^* = (C_1^*, C_2^*, C_3^*, v^*, w^*)$ and $C'^* = (C_1^*, C_2'^*, C_3^*, v^*, w^*)$, from the above queries, we can obtain $PK_s = aP$, $Q_s^* = h_1 = \gamma P$, $U = f_{s,t} = \gamma_1 P$, $U' = f_{s,t-1} = \gamma_2 bP$. $TK_0 = hk_0 P$, $TK_1 = hk_1 P$, $C_1^* = k_1 P$.

1) When $1 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \gamma P_{pub} + h \cdot \gamma_2 (hk_0 bP + abP) + U h h k_1 + U k_1.$$

$$C_2'^* = h' \gamma P_{pub} + h' \cdot \gamma_2 (hk_0 bP + abP) + U h' h k_1 + U k_1.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot P_{pub} + \gamma_2 (hk_0 bP + abP) + U h h k_1)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma P_{pub} - \gamma_2 h k_0 bP - U \cdot h k_1] \gamma_2^{-1} \gamma_2 bP.$$

2) When $0 \equiv t \pmod{2}$, $\mathcal{C}$ computes

$$C_2^* = h \cdot \gamma \cdot P_{pub} + h \cdot \gamma_2 (hk_1 bP + abP) + U \cdot h \cdot hk_0 + U \cdot k_1.$$

$$C_2'^* = h' \cdot \gamma \cdot P_{pub} + h' \cdot \gamma_2 (hk_1 bP + abP) + U \cdot h' \cdot hk_0 + U \cdot k_1.$$

Therefore, we can obtain

$$C_2^* - C_2'^* = (h - h')(\gamma \cdot P_{pub} + \gamma_2 (hk_1 bP + abP) + U \cdot h k_0)$$

Finally, $\mathcal{C}$ can compute CDHP:

$$abP = [(C_2^* - C_2'^*)(h - h')^{-1} - \gamma P_{pub} - \gamma_2 h k_1 bP - U \cdot h k_0] \gamma_2^{-1} \gamma_2 bP.$$

Resist side-channel attack: The proposed scheme can resist side-channel attacks and a certain degree of physical layer attacks. In the construction of the proposed scheme, a large number of random numbers are selected to generate partial private keys. By integrating the randomization mechanism, the physical signal in the signcryption process is unpredictable. In addition, we have added time periods to reduce differences in time or power consumption to defend against side channel attacks.

Resist physical layer attacks for RSU: Even if hardware tampering is carried out for RSU, the receiver of all information requires a digital signature to verify its origin and integrity. Therefore, the data tampering problem caused by physical layer attacks can be avoided to a certain extent.

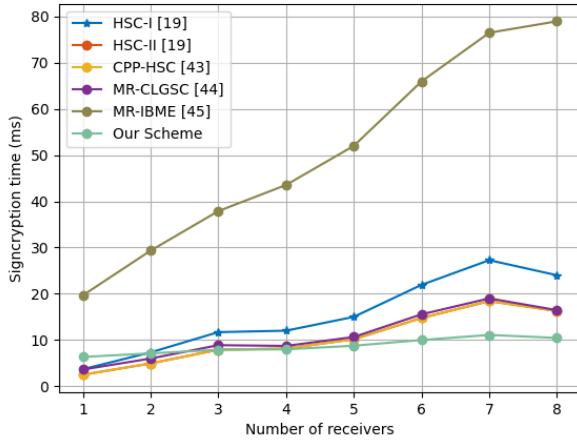
VI. PERFORMANCE ANALYSIS

During this section, the related works HSC-I [19], HSC-II [19], CPP-HSC [43], MR-CLGSC [44], MR-IBME [45] and our proposed scheme are evaluated. We demonstrate the performance of above schemes in terms of computing overhead, communication overhead, and functionality. We conduct the environment on the Windows 10 PC configured

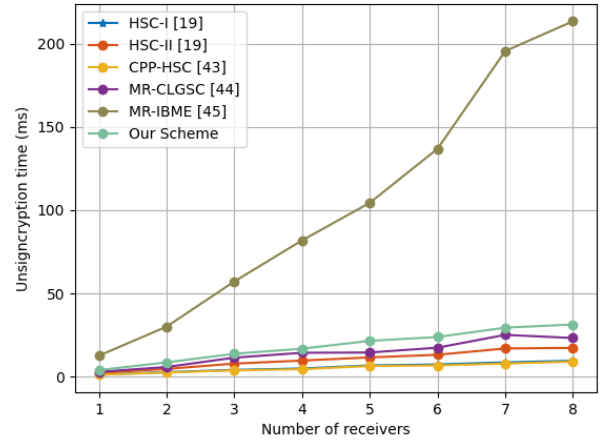
TABLE III: Comparison of efficiency and performance

Scheme		HSC-I [19]	HSC-II [19]	CPP-HSC [43]	MR-CLGSC [44]	MR-IBME [45]	Our scheme
Comp	SC/EC	$3nE_1+nE_2$	$2nE_1+nE_2$	$2nE_1+nE_2$	$(2n+1)E_1$	$3T_p+(2n+6)E_1+(3n+1)T_G$	$nT_p+3E_1+nE_2+T_G$
	USC/DEC	$2nT_p+nE_2$	$2nT_p+nE_1+nE_2$	$2nT_p$	$3nE_1$	$4nT_p+2n^2E_1+4nT_G$	$5nT_p+nE_1+2nE_2$
Comm	KG	E_1	E_1	*	*	E_1+T_G	$3E_1+3T_G$
	PK	$n G_1 $	$n G_1 $	$n G_1 $	$2n G_1 $	$n G_1 $	$n G_1 $
Function	CT	$3n G_1 $	$3n G_1 $	$3n G_1 +n Z_p^* $	$2 G_1 +(n+1) Z_p^* $	$(2n+4) G_1 +n Z_p^* $	$3 G_1 +n Z_p^* $
	HG	✓	✓	✓	×	×	✓
Security	MR	×	×	×	✓	✓	✓
	KI	×	×	×	×	×	✓
	DR	PKI-IBC	IBC-PKI	IBC-PKI	—	—	CLC-IBC
Assumption	IND-CCA2	IND-CCA2	IND-CCA2	IND-CCA2	IND-CCA2	IND-CCA2	IND-CCA
	EUF-CMA	EUF-CMA	EUF-CMA	EUF-CMA	EUF-CMA	EUF-CMA	EUF-CMA
Assumption	q-BDHIP	q-BDHIP	q-BDHIP	q-BDHIP	ECDLP	BDHP	BDHP
	&q-DHIP	&q-SDHP	&q-SDHP	&q-SDHP	&CDHP		&CDHP

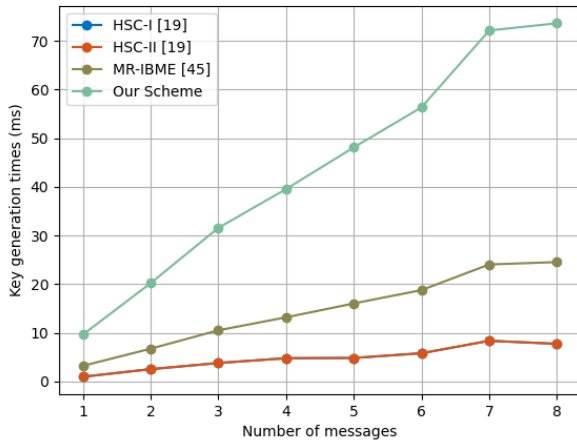
* Legends: Comp: the computational overhead, Comm: the communication overhead, SC: signcryption, USC: unsigncryption, KG: key generation of sender, CT: the size of ciphertext, PK: the size of public key, HG: heterogeneous signcryption, MR: multi-receiver, KI: key-insulated, DR: direction, q-BDHIP: q-bilinear Diffie-Hellman inversion problem, q-DHIP: q-Diffie-Hellman inversion problem, q-SDHP: q-Strong Diffie-Hellman problem, ECDLP: elliptic curve discrete logarithm problem, CDHP: computational Diffie-Hellman problem, BDHP: bilinear Diffie-Hellman problem, *: negligible, —: non-existent, ✓: satisfy this function, ×: cannot satisfy this function.



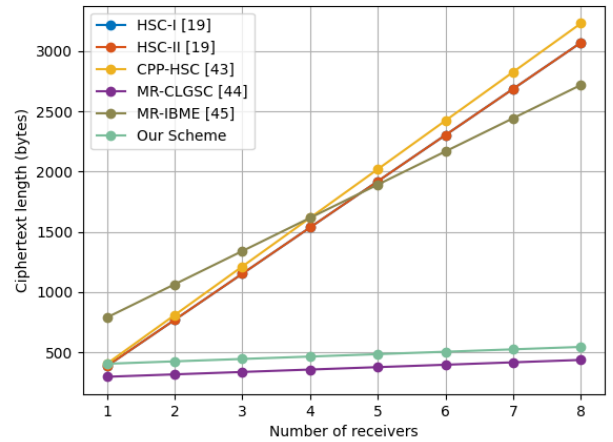
(a) The computation overhead of signcryption



(b) The computation overhead of unsigncryption



(c) The computation overhead of key generation



(d) The communication overhead of ciphertext

Fig. 8: The computation and communication overhead of different schemes.

TABLE IV: The analysis of the advantages and disadvantages of compared schemes

Scheme	CF	UF	AVKL	MR	HG	DR	Advantages	Disadvantages
HSC-I [19]	✓	✓	×	×	✓	PKI → IBC	Achieve confidentiality unforgeability and heterogeneity	Cannot avoid key leakage One-to-many transmission High signcryption overhead
HSC-II [19]	✓	✓	×	×	✓	IBC → PKI	Achieve confidentiality unforgeability and heterogeneity	Cannot avoid key leakage One-to-many transmission High signcryption overhead
CPP-HSC [43]	✓	✓	×	×	✓	IBC → PKI	Achieve confidentiality unforgeability and heterogeneity Batch verification	Cannot avoid key leakage One-to-many transmission
MR-CLGSC [44]	✓	✓	×	✓	×	—	Achieve confidentiality Unforgeability and one-to-many transmission	Cannot avoid key leakage and heterogeneity
MR-IBME [45]	✓	×	×	✓	×	—	Achieve confidentiality Revocable encryption One-to-many delivery	Cannot avoid key leakage Unforgeability and heterogeneity
Our scheme	✓	✓	✓	✓	✓	CLC → IBC	Achieve the listed function Low computational overhead	Cannot batch verification

* Legends: CF: confidentiality, UF: unforgeability, AVKL: avoid key leakage.

with i7-11700F CPU @2.50GHz and 16 GB memory. The supersingular curve $E/F_p : y^2 = x^3 + x$ over a finite field F_p is employed to achieve the 1024-bits RSA. The cryptographic operations such as T_p , E_1 , E_2 and T_G are given, where T_p represents the bilinear pair running time, E_1 represents the scalar multiplication on G_1 , E_2 represents the exponentiation operation on G_2 , and T_G represents the operation mapped from hash to G_1 . We randomly select three experiment results and demonstrate the results in Table II. In addition, we can obtain that $|Z_p^*| = 20$ bytes and $|G_1| = |G_2| = 128$ bytes.

A. Comparison of Computational Overhead

In this section, we demonstrate computational overhead comparison of competitive works, and concrete comparison results are shown in Table III. In particular, if there exist n receiving users in the system, then the signcryption operation in scheme [43] requires $2n$ scalar multiplications on G_1 and n exponentiation operations on G_2 . Therefore, the signcryption overhead in [43] is $2nE_1 + nE_2$. The same method of signcryption calculation is also employed in other schemes. Besides, with the increased number of receivers, the changing trend of signcryption overhead is described in Fig. 8(a), we observe that if $n < 3$, the scheme in [43] has a lower overhead than other competitive schemes. Nonetheless, if $n \geq 3$, our proposed scheme has a lower overhead than other comparative schemes. In addition, the unsigncryption operation in scheme [43] requires $2n$ bilinear pair running times, which is defined as $2nT_p$.

With the increased number of receivers, the cost trend of unsigncryption is demonstrated in Fig. 8(b). We find that the overhead of our construction is lower than schemes in [45]. The key generation operation of sender in scheme [45] requires one scalar multiplication on G_1 and one hash to G_1 operations, which is computed as $E_1 + T_G$. It should be noted that the key generation costs of schemes in [43] and [44] are negligible compared with other schemes. In key generation phase, such as Fig. 8(c), our proposed HPKI-MRSC is more expensive than other works, it is forgivable since that parallel key-insulated mechanism is integrated at this stage, and the changing helper keys occupy a certain amount of time overhead.

B. Comparison of Communication Overhead

In this section, we demonstrate communication overhead of competitive works, and concrete comparison results are also shown in Table III. From Table III, we can obtain that the public key length utilized in signcryption of scheme [44] is $2n|G_1|$, where the other schemes require $n|G_1|$ overhead. The ciphertext communication overhead changes with the increased number of receivers as shown in Fig. 8(d). The communication cost of our proposed scheme is slightly more expensive than that in scheme [44]. Nonetheless, the ciphertext generated by our construction is shorter than other competitive schemes. Therefore, in the actual transmission of ciphertext, the time cost of proposed HPKI-MRSC is more suitable for lightweight equipment transmission in IoV with the same level of security.

C. Comparison Analysis of Functionality

In this section, we evaluate the functionality of our scheme with other competitive schemes, the comparison results are also demonstrated in Table III and Table IV. Obviously, we obtain that all the listed schemes have heterogeneous function with the exception of schemes [44] and [45]. However, our scheme and schemes in [44], [45] can support the multiple receivers. This mechanism can greatly increase the efficiency of information transmission and reduce the transmission overhead of ciphertext. As for the key-insulated mechanism, only our proposed scheme can support this function, and sender's private key can be constantly updated during time period to resist key disclosure threat. In summary, the proposed HPKI-MRSC scheme can satisfy all the functions listed in Table IV.

VII. CONCLUSION

In this paper, we proposed a heterogeneous parallel key-insulated multi-receiver signcryption scheme for IoV, to provide a message transmission solution between CLC and IBC system. The integrated multi-receiver mechanism is designed to enable one sender to send message to multiple receivers, hence substantially improving the efficiency of message transmission of V2V and V2I interactions in IoV systems.

Besides, the parallel key-insulated mechanism is employed in our construction to mitigate the risks of private key leakage in the deployment environment. Finally, we demonstrated security analysis and performance analysis of the proposed scheme, and the result indicates that our proposed HPKI-MRSC is more suitable for IoV environment. Future work will consider the aggregation signcryption idea to improve the validation efficiency of messages. Besides, we will add the feasibility and effectiveness of the scheme in actual IoV environments [46], [47]. Based on this, we will explore the rationality and feasibility of different assumptions in practical applications [48], [49].

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